



Government debt, income distribution and growth

Jong-Il You* and Amitava Krishna Dutt†

This paper takes up the question of whether government debt worsens the distribution of income as tax revenues from workers are used to pay for the interest on the debt held by the rich. In so doing, we develop a post-Keynesian model in which growth is determined by aggregate demand rather than by the supply of resources and income is distributed between workers who earn wages and capitalists who receive profit and interest income. Our analysis highlights the possible expansionary effect of a rise in government debt which may raise the income of the workers, and it shows that its precise effect on income inequality depends on the circumstances under which government debt rises.

© 1996 Academic Press Limited

1. Introduction

Three prominent facts regarding the US economy in the 1980s were a rapid build-up of government debt, consistently low investment rates and productivity growth rates, and a significant deterioration in the distribution of income. Government debt as a percentage of GDP increased from 26.3 in 1980 to 44 in 1990.¹ Gross domestic investment as a ratio of GDP fell from 18% in 1980 to 16% in 1990.² Productivity growth has been about 1.2% in the 1970s and 1980s compared with 2.8% in the 1950s and 1960s.³ Regarding income distribution, the percentage of income going to the lowest quintile (under \$15,102 in 1988) fell from 5.1 in 1980 to 4.6 in 1988, while that going to the highest (over \$55,906 in 1988) increased from 41.6 to 44.0. Average family income of the lowest income decile fell 1.48% between 1977 and 1988 in 1987 dollars, while that for the highest increased by 16.5%. The incomes of the lower eight deciles actually fell, while that of the top 1% increased by 49.8% over the period.⁴

Many analysts believe that these developments are not unrelated. Most mainstream economists as well as politicians see a linkage between the deficits and low growth owing to

Manuscript received 27 September 1993; final version received 8 June 1994.

*Ritsumeikan University, Japan and †University of Notre Dame, USA.

The order in which the names appear is random. We wish to thank the anonymous referees for this journal for their helpful comments on an earlier draft.

¹ Debt-GDP ratio is taken from DRI/McGraw Hill, cited in Dornbusch and Fischer (1994).

² These figures are taken from *World Development Reports* for 1982 and 1992. The corresponding ratios in 1990 were 33 for Japan and 22 for Germany; the US figure is the lowest among the high-income economies according to the World Bank classification.

³ The 1970 and 1980 figures are considerably lower than those of Japan and Germany, though these countries also experienced a slowdown in productivity growth. See Krugman (1990).

⁴ These figures are taken from US Census Bureau and the Urban Institute's *Challenge to Leadership*, and are cited in Phillips (1990, pp. 13, 17).

the crowding out of private investment by rising government deficits, and slow growth in turn is linked to poor productivity performance. Thus, deficit reduction is seen as a key to restoring investment and ensuring long-term regeneration of growth. Some on the left of the political spectrum have jumped on the bandwagon against deficits, but on entirely different grounds. They argue that tax revenues from the working class are used to pay for the interest on the government debt which is owned almost exclusively by a small minority of wealthy individuals, so that government debt contributes to income inequality.¹

While the connection between the debt, and investment and growth has received much attention in the literature,² the connection between debt and income distribution mentioned above has largely escaped serious economic analysis. This paper takes some steps towards filling this serious lacuna using a formal model of distribution, growth and government debt.

While the worsening income distribution in the US has attracted quite a bit of attention recently, the prevailing analysis focuses on the increase in wage inequality associated with developments within the labour market (for example, OECD, 1993 and Blackburn *et al.*, 1990). But there are other forces at work. Glyn (1992) shows that the share of the profit income out of the produced income (GDP) increased in most advanced countries including the US during the 1980s. In addition, the interest (non-produced) income of the private sector on its holdings of the government debt rose steeply: the net interest paid by the US Federal Government rose from \$52.7 billion or 1.9% of GDP in 1980 to \$176.6 billion or 3.2% of GDP in 1990.³ Concentration of the profit income is well known, and Michl (1991) shows that the top 20% of the income class receives over 86% of the direct interest payments on Treasury issues and about 75% of the total (that is, direct and indirect) interest paid by the government.

Thus, it is apparent that growing government debt could have been an important factor worsening the distribution of income. But it is incorrect to argue, without further analysis, that growing government debt causes a worsening of distribution. Even if it is true that the growth of the government debt is skewing income distribution it is not at all clear whether distribution would be less unequal if the debt had not grown. To examine this issue one has to be clear about what caused the rise in the government debt and what would have happened had these causes not been operative. To analyse this, one has to go beyond impressionistic arguments and examine a rigorous theoretical model.

The model developed in this paper is a post-Keynesian one in the sense that it takes the growth rate of the economy to be constrained by aggregate demand rather than resource supplies, that it takes demand to depend largely on the animal spirits of firms, and that it takes money supply to be determined by demand. Elsewhere, we have used such a post-Keynesian model to examine the conventional argument against debt relating to crowding out (see Dutt and You, 1993). In this paper we modify that post-Keynesian model to analyse distributional issues by introducing into it another important characteristic of post-Keynesian economics: differential propensities to save out of wage and non-wage income.

There are some earlier examples of growth models in the post-Keynesian literature in which income distribution and government debt play important roles. Pasinetti (1989)

¹ See, for example, the argument by Heilbroner and Bernstein (1989, p. 50) and National Economic Commission (1989, p. 56), Michl (1991), Baldani and Michl (1987).

² See, for instance, Bernstein (1989), Eisner (1989), Heilbroner and Bernstein (1989), several papers in Rock (1991), and Dutt and You (1993).

³ These figures are based on the data reported in the *Economic Report of the President*, 1993.

discusses the implications of different propensities to consume out of wage and non-wage income in a dynamic model with government debt and deficits. However, his analysis is confined to the case of an economy growing at the natural rate with fully employed labour for which other assumptions for Ricardian equivalence are satisfied. Ewijk (1991) has developed a model which is more post-Keynesian in nature, since growth is determined by investment and consumption demand. However, since he assumes that output is at full capacity in the dynamic process in which the accumulation of debt is examined, the role of demand is muted in his long-run analysis, pace the post-Keynesian approach.¹ Moreover, Ewijk assumes that capitalists are net borrowers, so that all interest income—including that on government bonds—is received by workers, which is at odds with the complaint mentioned above that the existence of debt results in interest payments which worsen the income distribution by redistributing income away from workers.

The remainder of the paper is organised as follows: Section 2 examines the model for the short run; Section 3 analyses the model for the long run in which stocks of capital and debt change over time according to investment by firms and borrowing by the government; Section 4 examines factors that cause a rise in the debt–capital ratio; Section 5 pulls together the analysis of the earlier sections to examine the relationship between government debt and income distribution; Section 6 concludes.

2. The short-run model

This section examines the structure and behaviour of the model for the short run in which the stocks of capital and government debt are assumed to be given and the market for goods clears owing to variations in output. We first consider the determinants of output and growth and then consider income distribution.

2.1. Determination of output and growth

We consider a closed economy where the output (X) is determined by the aggregate demand; labour is in unlimited supply and firms produce under conditions of excess capacity.

Assuming that workers do not save, while capitalists save a fraction, s , of their disposable income, consumption demand is

$$C = (1 - \tau_w)(1 - \pi)X + (1 - s)(1 - \tau_c)(\pi X + iD/P), \quad (1)$$

where π is the profit share of income, τ_w and τ_c the income tax rates for the wage income and the capital income, i the nominal interest rate on the government debt, D the stock of government debt in nominal terms, and P the price level.² We shall assume throughout

¹ Ewijk's analysis also departs from standard post-Keynesian tradition in determining the interest along standard *IS-LM* lines and does not allow for money supply endogeneity in the post-Keynesian sense. Moreover, he tries to incorporate too many factors—like alternative government policy rules, and savings by workers—which forces him to resort to simulation techniques in his analysis of long-run debt dynamics.

² If we allow for workers' saving, we have to add at least one more state variable Θ , the proportion the government debt held by capitalists, in the dynamic analysis. Owing to the non-linearities in the dynamic model, this will enormously complicate the analysis. However, the central feature of our model (see below)—that a rise in the debt increases capacity utilisation and the inequality ratio—will be preserved as long as $\Theta > \pi$. Supposing that the share of the total interest paid by the government received by the top 20% of the income class is a reasonable approximation of Θ , $\Theta \approx 0.75$ according to Michl (1991) which far exceeds $\pi \approx 0.25$, a figure taken from Dornbusch and Fischer (1994).

this paper that π , i and P are exogenously given.¹ Note that the pre-tax capital income includes not only the profits from current production but the interest income paid by the government.

Investment demand depends on the expectations of firms regarding profits and market prospects. Noting that the current after-tax rate of profit is given by $r = (1 - \tau_c)\pi(X/K)$ where K is the stock of capital, we write the investment function in the simple linear form

$$I = [a_0 + a_1 u + a_2(1 - \tau_c)\pi]K, \quad (2)$$

where $u = X/K$ is the rate of capacity utilisation.² The rate of interest as a determinant of investment demand is ignored for now, given the assumption that it is fixed. In assuming that the interest rate is fixed we are following a frequent practice in the post-Keynesian literature which assumes that money supply is endogenously determined by money demand at a given interest rate.

Finally, we assume

$$G = \gamma K, \quad (3)$$

that is, the real government expenditure is maintained as a constant fraction of the capital stock.

Collecting (1–3), we have in the short-run equilibrium in which the goods market clears through variations of output,

$$X = (1 - \tau_w)(1 - \pi)X + (1 - s)(1 - \tau_c)(\pi X + iD/P) + [a_0 + a_1 u + a_2(1 - \tau_c)\pi]K + \gamma K. \quad (4)$$

Defining $\delta = D/PK$, the ratio of the real government debt to the capital stock, dividing both sides of (4) by K and solving for u , we obtain the short-run equilibrium capacity utilisation rate:

$$u(\delta, \pi) = \lambda(\pi)[(1 - s)(1 - \tau_c)i\delta + a_0 + a_2(1 - \tau_c)\pi + \gamma], \quad (5)$$

where $\lambda(\pi) = (\tau_w + \sigma\pi - a_1)^{-1}$, with $\sigma = (1 - \tau_w) - (1 - s)(1 - \tau_c)$, is the Keynesian output multiplier. We assume both $\lambda > 0$ and $\sigma > 0$.

Two observations are in order. First, the stock of the government debt has a positive impact on output, since the interest payment by the government stimulates consumption demand by raising the disposable income of the capitalists. This can be seen from

$$\partial u / \partial \delta \equiv \Lambda(\pi) = \lambda(\pi)(1 - s)(1 - \tau_c)i > 0. \quad (6)$$

Second, the distribution of income between profits and wages has an ambiguous effect on aggregate demand, since a rise in the profit share reduces consumption demand but may increase investment demand. From

¹ The assumption regarding the fixity of π can be taken to be based on that of a fixed markup set by firms on prime costs à la Kalecki. If the markup is z we have $\pi = z/(1 + z)$.

² In the post-Keynesian literature the following specification of investment function is frequently used (see Rowthorn, 1982; Dutt, 1984):

$$I = (a_0 + a_1 u + a_2 r)K.$$

This equation directly captures the two influences of expectations regarding markets and profitability, where expectations are equal to current values. While the formulation in the text allows possibilities of a rise in the profit share having a positive or a negative effect on capacity utilisation, the above formulation allows only the negative effect. As such, the investment function used in the text, which follows Bhaduri and Marglin (1990), is more general. This generality allows our analysis to consider the implications of a variety of influences which can possibly make a rise in a profit share increase capacity utilisation—such as savings by workers and firms, overhead labour, and open economy considerations (see Steindl, 1952; Blecker, 1989; Taylor, 1991; and Lavoie, 1992)—without explicitly introducing them into our model.

$$\partial u / \partial \pi = [\alpha_2(1 - \tau_c) - \sigma u] \lambda(\pi), \quad (7)$$

noting that $\partial(I/K)/\partial\pi = \alpha_2(1 - \tau_c)$ and $\partial(C/K)/\partial\pi = -\sigma u$ (where the partial derivatives are taken holding u constant), we see that a rise in the profit share will be expansionary if the positive effect on investment demand is greater than the negative effect on consumption demand; we call this the case of profit-led expansion. But if the positive effect on investment demand is smaller than the negative effect on consumption demand, a rise in the profit share will be contractionary; this case, in which a rise in the wage share will increase capacity utilisation, we shall call the case of wage-led expansion.¹

The rate of capital accumulation in the economy can now be determined by the level of investment at the short-run equilibrium level of capacity utilisation. Assuming away the depreciation of capital, this is given by

$$g(\delta, \pi) = I/K = \alpha_0 + \alpha_1 u(\delta, \pi) + \alpha_2(1 - \tau_c)\pi. \quad (8)$$

Since

$$\partial g / \partial \delta = \alpha_1 \Lambda(\pi) > 0,$$

the rate of accumulation rises with δ : the rise in δ increases capacity utilisation and, hence, through the accelerator effect, the rate of investment and capital accumulation. Since

$$\partial g / \partial \pi = \alpha_1(\partial u / \partial \pi) + \alpha_2(1 - \tau_c),$$

the effect of a rise in π on the rate of accumulation is ambiguous. If we have the case of profit-led expansion, g must increase with π ; we shall call this the case of profit-led growth. Even if we have wage-led expansion and the first term is negative, since the second term is positive it is still possible to have g rise with π and thus have profit-led growth. However, if we have wage-led expansion and the first term is so strongly negative as to make the whole expression negative, g will rise when π falls, when we have the case of wage-led growth.

2.2. Income distribution

Given the simple two-class assumption, the obvious measure of income inequality is that of the class distribution of income. We define the inequality ratio as the ratio of the capitalists' income (capital income) to the workers' income (labour income), both after-tax,

$$v(\delta, \pi) = (1 - \tau_c)(\pi u + i\delta) / (1 - \tau_w)(1 - \pi)u.$$

This can be rewritten as

$$v(\delta, \pi) = q(\pi + i\delta/u) / (1 - \pi), \quad (9)$$

where $q \equiv (1 - \tau_c) / (1 - \tau_w)$. There are three classes of determinants of this ratio: the relative tax rates (q), the profit share of the income from production or output (π) and the ratio of the interest income to the income from production ($i\delta/u$), though the last is not independent of the first two.

We now consider the determinants of income distribution in the short run in which δ is given. First, when the tax rates are altered, the inequality ratio is affected in two separate ways. One is through its effect on q , whose direction is straightforward. For later

¹ Bhaduri and Marglin (1990) refer to these cases as the exhilarationist and stagnationist cases. We prefer expressions which do not evoke positive or negative emotions regarding the cases.

reference, we define a 'fair' tax hike or cut as a change in tax rates that leaves q unaffected, that is, $(1 - \tau_w)d\tau_c = (1 - \tau_c)d\tau_w$. The other way in which a change in the tax rates affect income distribution is through its effect on u . A 'fair' tax hike will worsen income distribution in our model, because it reduces the income from production which is divided between the capitalists and the workers in a fixed proportion while the interest income that goes only to the capitalists remains unchanged. (A cut in government spending or a collapse of 'animal spirits' leading to a fall in α_0 will also have a regressive effect on income distribution for exactly the same reason.)

A rise in the profit share of output may have a perverse effect on the inequality ratio in the short run. Its direct effect is to increase inequality, but in the profit-led case if it raises output by a large amount it is possible for the inequality ratio to go down. From

$$\partial v / \partial \pi = [v + q - q(i\delta/u^2)(\partial u / \partial \pi)] / (1 - \pi),$$

a rise in π could reduce v in the case of profit-led expansion if $\partial u / \partial \pi$ is very large. However, since the interest income as a share of the output ($i\delta/u$) is usually small, this perverse case is highly unlikely to arise.

It is obvious from (8) that the government debt is also a determinant of the income inequality. However, δ cannot vary in the short run. The issues regarding the government debt and the distribution of income, therefore, will be analysed later in the context of the long-run dynamics. Note also that the short-run effect on the inequality ratio of the various exogenous shocks considered above would not in general be a reliable indicator of what is going to happen in the long run, because they will affect the evolution of δ and thereby have further consequences on v . We shall examine these issues in Section 4.

3. The long-run model: dynamics of the government debt

In this section we analyse the dynamics of the government debt and the stock of capital. The long-run equilibrium is defined as one in which the debt-capital ratio δ remains constant over time.

For the sake of simplicity we ignore the monetary assets and assume that the entire deficit has to be financed by issuing debt:

$$dD/dt = P(G - T) + iD, \quad (10)$$

where $T = \tau_w(1 - \pi)X + \tau_c(\pi X + iD/P)$ is the real government revenue consisting solely in income taxes.¹ Differentiating $\delta = D/PK$ with respect to time and using (10), we obtain the familiar equation of motion for the government debt:

$$d\delta/dt = [\gamma - \tau^*u(\delta) - \tau_c i\delta] + [i - g(\delta)]\delta, \quad (11)$$

where $\tau^* = \tau_w(1 - \pi) + \tau_c \pi$ is the average income tax rate. It shows that the evolution of the debt-to-capital ratio is governed by the primary deficit (the first bracketed term) and the difference between the interest rate (i would have to be replaced by the real interest rate if the fix-price assumption is relaxed) and the growth rate. Note that short-run equilibrium values of u and g are expressed as functions of δ only, reflecting the assumption that π is fixed. Using

¹ It is not difficult to introduce monetisation of debt in the model. Since this will not alter the results of our paper in any significant way we eschew this complication here. But see Dutt and You (1993).

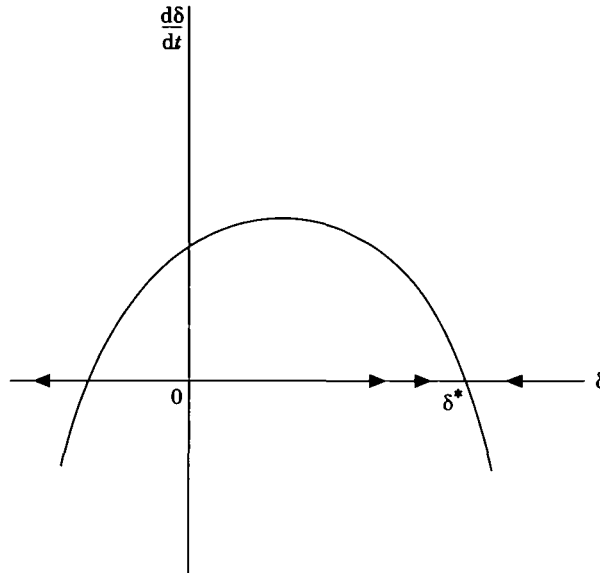


Fig. 1. Stability of the long-run equilibrium debt-capital ratio.

$$u(\delta) = \lambda[\alpha_0 + \alpha_2(1 - \tau_c)\pi + \gamma] + \Lambda\delta = u(0) + \Lambda\delta \quad (12)$$

$$g(\delta) = [\alpha_0 + \alpha_1 u(0) + \alpha_2(1 - \tau_c)\pi] + \alpha_1 \Lambda\delta = g(0) + \alpha_1 \Lambda\delta,$$

equation (11) can be rearranged as

$$d\delta/dt = a_1\delta^2 + a_2\delta + a_3, \quad (13)$$

where $a_1 = -\alpha_1\Lambda$, $a_2 = (1 - \tau_c)i - g(0) - \Lambda\tau^*$ and $a_3 = \gamma - \tau^*u(0)$.

Provided that $a_2^2 - 4a_1a_3 > 0$, there are two solutions for the long-run equilibrium ($d\delta/dt = 0$). Since $a_1 < 0$, the greater of them,

$$\delta^* = (-a_2 - \sqrt{a_2^2 - 4a_1a_3})/2a_1 \quad (14)$$

is the unique stable equilibrium (see Fig. 1).¹ Therefore, if δ is currently increasing, it will necessarily reach δ^* and stabilise. Otherwise, there is no need to worry about the debt in the first place. We shall assume for the remainder of the paper that $a_3 = \gamma - \tau^*u(0) > 0$, which means that there would be a primary deficit when $\delta = 0$. This ensures not only the existence of the stable equilibrium but that δ^* is positive.

As we have discussed elsewhere (Dutt and You, 1993), the local stability condition for the equilibrium, $\partial(d\delta/dt)/\partial\delta < 0$, is satisfied at δ^* owing to the fact that

$$\partial^2(d\delta/dt)/\partial\delta^2 = 2a_1 = -\alpha_1\Lambda < 0.$$

Recalling that $\alpha_1 = \partial g/\partial u$ and $\Lambda = \partial u/\partial\delta$, we can see that the demand stimulating effect of the debt and the interaction between the multiplier and the accelerator effect are

¹ If $a_2^2 - 4a_1a_3 < 0$, there is no stationary equilibrium and δ will be decreasing forever (i.e., the government keeps on accumulating financial claims on the private sector without limit). Clearly, this is an unsustainable process and such a policy regime not worthy of consideration. If $a_2^2 - 4a_1a_3 = 0$, there exists an equilibrium at $\delta^* = -a_2/2a_1$. But it is unstable, and a slight perturbation would lead the economy on to the above unsustainable path.

responsible for the stability result.¹ Therefore, it is crucial that the economy does not hit the capacity constraint in this process.²

In the long-run equilibrium in which the debt–capital ratio remains at δ^* , the growth rate of output as well as the inequality ratio also remain constant at their long-run equilibrium values. From equations (12), the long-run equilibrium growth rate is

$$g^* = g(\delta^*) = (1 + a_1\lambda)[a_0 + a_2(1 - \tau_c)\pi] + a_1\lambda\gamma + a_1\Lambda\delta^*, \quad (15)$$

noting that in the long-run equilibrium the rate of capital accumulation is the same as the rate of output growth since u remains the same. The long-run equilibrium value of the inequality ratio is, from (9),

$$v^* \equiv v(\delta^*) = q[\pi + i\delta^*/u(\delta^*)]/(1 - \pi), \quad (16)$$

where $u(\delta^*) = \lambda[a_0 + a_2(1 - \tau_c)\pi + \Lambda\delta^*]$ from (12).

While the economy converges to its long-run equilibrium, with no changes in the parameters during the out-of-equilibrium dynamics, we can see how g and v evolve as δ rises by taking partial derivatives of (15) and (16). That g should be rising was shown above, while, from (16),

$$\partial v / \partial \delta = qi(u - \Lambda\delta)/(1 - \pi)u^2 = qiu(0)/(1 - \pi)u^2 > 0$$

shows that v must also be rising. As δ increases, it will raise the interest income directly, causing v to increase, and the produced income by stimulating the aggregate demand, causing v to decrease. The above expression shows that the former effect dominates and the inequality ratio increases along the path toward the long-run equilibrium.

It may be objected that we have assumed away the problems of inflation and crowding out due to rising interest rate which present obstacles to the debt-led expansion. While these are undoubtedly important concerns, we believe that the fears of such problems are often exaggerated (see Dutt and You, 1993) and that, given the persistently high unemployment in much of the world, our starkly post-Keynesian assumptions provide at least a good starting point.

4. Causes of a rise in the debt–capital ratio

In this section we identify possible causes of a rise in δ^* by analysing how policy changes and other shocks to the economy will affect the long-run equilibrium debt–capital ratio, growth rate and the inequality ratio. To do so, it is convenient to have the general expression for the change in the long-run equilibrium debt–capital ratio. Totally differentiating the identity

$$a_1\delta^{*2} + a_2\delta^* + a_3 = 0,$$

we get

¹ It may be noted that the government spending as a proportion of GDP decreases as δ increases. Since $G/X = \gamma K/X = \gamma/u(\delta)$ and γ is fixed,

$$\partial(G/X)/\partial\delta = -\Lambda/u^2 < 0.$$

But this is not an essential mechanism behind the stability result. Changing the government spending policy to $G = \gamma X$ will not alter the stability result, as a_1 would still be negative.

² It may be objected that there should not be excess capacity in the long run. Following Kalecki, we conceptualise the long run as a succession of short runs instead of a fictitious time frame in which all plans are realised and ‘full’ capacity utilisation prevails.

$$d\delta^* = [(da_1)\delta^{*2} + (da_2)\delta^* + (da_3)]/\sqrt{a_2^2 - 4a_1a_3}, \quad (17)$$

where we use $(2a_1\delta^* + a_2) = -\sqrt{a_2^2 - 4a_1a_3}$ from (14).

4.1. Expansionary fiscal policies

Suppose the government attempts a fiscal expansion by increasing government spending, that is, by raising γ . Normally, δ^* should increase following a rise in γ . Surprisingly, however, this is not necessarily the case. Since

$$\partial a_1/\partial \gamma = 0, \partial a_2/\partial \gamma = -\alpha_1\lambda \text{ and } \partial a_3/\partial \gamma = 1 - \tau^*\lambda,$$

we can see from (17) that a counter-intuitive result of a fall in δ^* will obtain if

$$\lambda > 1/(\tau^* + \alpha_1\delta^*),$$

that is, if the multiplier is large. We can explain this result as follows. If the multiplier is larger than the inverse of the average tax rate ($\lambda > 1/\tau^*$), a rise in government spending will actually decrease the deficit and hence the debt since the tax revenues will rise by more than the spending.¹ As we can see from (11), moreover, δ may decrease even if the current deficit (dD/dt) rises owing to the growth of capital, given that δ^* is positive. Therefore, the condition for a rise in γ to reduce δ^* is less stringent than the condition for it to decrease the deficit in the short run. However, a back-of-the-envelope calculation with plausible numbers suggests that it is not likely to be met unless δ^* is extremely large.

Supposing that a rise in γ does lead to an increase in δ^* , the long-run growth rate will definitely increase. In this case it results in an increase in capacity utilisation, which through the accelerator effect on investment increases the growth rate, because of both the direct injection of the government demand and the increase in the consumption demand due to the rise in δ^* . Since, from (6) and (15),

$$dg^* = \alpha_1\lambda[d\gamma + i(1-s)(1-\tau_c)(d\delta^*)]$$

and $i(1-s)(1-\tau_c)$ is 'small', even if δ^* falls the growth rate is likely to increase.

The effect on the long-run distribution is ambiguous if δ^* rises. Noting $du^*/d\gamma = \lambda + \Lambda(d\delta^*/d\gamma)$, totally differentiating (16) and using (12), we obtain

$$dv^* = qi[u(0)(d\delta^*) - \lambda\delta^*(d\gamma)]/(1-\pi)u^2,$$

which shows the contradictory effects of a rise in γ on the inequality ratio in this case. The rise in u^* directly caused by the fiscal expansion tends to reduce the inequality ratio, countering the indirect effect of rising δ^* which raises the inequality ratio. As $u(0)/\lambda\delta^*$ should be fairly large, worsening income distribution is quite possible in this case. Clearly, if the fiscal expansion leads to a fall in δ^* , the inequality ratio must also fall.

We can summarise the above results in terms of the value of $d\delta^*/d\gamma$ when γ rises:

- (i) If $d\delta^*/d\gamma > \lambda\delta^*/u(0)$, δ^* , g^* and v^* will all increase.
- (ii) If $0 < d\delta^*/d\gamma < \lambda\delta^*/u(0)$, both δ^* and g^* increase but v^* falls.
- (ii) If $-[i(1-s)(1-\tau_c)]^{-1} < d\delta^*/d\gamma < 0$, both δ^* and v^* fall but g^* still rises.
- (iv) If $d\delta^*/d\gamma < -[i(1-s)(1-\tau_c)]^{-1}$, δ^* , g^* and v^* all decrease.

¹ The traditional short-run Keynesian stability condition that the total saving must increase by more than investment demand when output rises puts no restriction on λ other than it must be positive. Denoting private saving by S , the short-run stability condition is

$$\partial(T+S)/\partial X - \partial I/\partial X = \tau^* + s(1-\tau_c)\pi - \alpha_1 = 1/\lambda > 0.$$

If, as is frequently assumed, the marginal *private* saving propensity is higher than the accelerator effect ($s(1-\tau_c)\pi > \alpha_1$), then $\lambda < 1/\tau^*(\partial a_3/\partial \gamma > 0)$ and this kind of anomaly cannot happen.

Outcome (i), a rise in the debt-capital ratio and the growth rate and a deterioration of income distribution, is the result one would expect. Outcome (ii) is also likely, so that the effect on income distribution is genuinely ambiguous. Outcome (iii) is less likely but of interest since it points to an intriguing and blissful possibility of reducing the debt-capital ratio, increasing the growth rate and improving income distribution by a single stroke of fiscal expansion. Outcome (iv) is extremely unlikely.

The long-run effects of a fiscal expansion through a tax cut are essentially similar to the case of a spending hike, and the conclusions drawn above would not change in any significant way. The effect on income distribution is ambiguous, and there exists a possibility of a fall in δ^* and g^* . However, there is one minor difference. Since a cut in τ_c stimulates investment by raising the after-tax profit rate in addition to the expansionary effect of the rise in deficits, the perverse possibility of g^* going down is even more remote than in the case of a spending hike while, by the same token, the possibility of δ^* going down is enhanced.

4.2. Autonomous slowdown of growth

Suppose some adverse shocks such as a collapse of the 'animal spirits' cause a fall in autonomous investment (α_0). The ambiguities that arose in our analysis of the fiscal policy are largely absent in this case. As the rate of capital accumulation falls, the debt-capital ratio unambiguously rises. Since

$$\partial a_1 / \partial \alpha_0 = 0, \partial a_2 / \partial \alpha_0 = -(1 + \alpha_1 \lambda) < 0 \text{ and } \partial a_3 / \partial \alpha_0 = -\tau^* \lambda < 0,$$

from (17), we know that δ^* will increase when α_0 falls. Moreover, the build-up of debt in this case is unambiguously associated with worsening income distribution. Since the fall in α_0 has an adverse effect on output for any given δ in addition to raising δ^* , the inequality ratio must increase as can be confirmed from

$$dv^* = q[u(0)(d\delta^*) + \lambda \delta^* (-d\alpha_0)] / (1 - \pi) u^2 > 0.$$

The effect on the long-run equilibrium growth rate is ambiguous, however, since the direct effect of the autonomous investment slowdown is countered by a rise in δ^* in the long run which increases the induced investment. From

$$dg^* = -(1 + \alpha_1 \lambda)(-d\alpha_0) + \alpha_1 \Lambda(d\delta^*)$$

and the fact that $\alpha_1 \Lambda$ is small, however, we can see that the growth rate will decline unless the increase in δ^* is extremely large.

4.3. Rise in the profit share

A rise in π reflecting an increase in the 'monopoly power' of the firms can be caused by a rise in industrial concentration or a decline in the bargaining power of workers. It is useful to start by identifying the short-run effects (that is, with a given δ) of a rise in π . Firstly, it has an adverse effect on the average tax rate, which may be either positive or negative depending on the progressivity of the tax system ($\partial \tau^* / \partial \pi = \tau_c - \tau_w$). Secondly, the magnitude of the output multiplier will be reduced, since the shift in income distribution toward profits in effect reduces the economy-wide marginal propensity to consume ($\partial \lambda / \partial \pi = -\sigma \lambda^2 < 0$). Finally, a rise in π has an effect on capacity utilisation, which is ambiguous as noted in (7), and thereby influences the capital accumulation rate [$\partial g / \partial \pi = \alpha_2(1 - \tau_c) + \alpha_1(\partial u / \partial \pi)$].

Keeping the above in mind and using (17), we can obtain

$$d\delta^*/d\pi = -[(\tau_c - \tau_w)u^* + (\partial u/\partial \pi)(\tau^* + a_1\delta^*) + a_2(1 - \tau_c)\delta^*]/\sqrt{a_2^2 - 4a_1a_3}.$$

Therefore, if the capital income tax rate is higher than the labour income tax rate ($\tau_c > \tau_w$) and the economy satisfies the case of profit-led expansion ($\partial u/\partial \pi > 0$), the debt-capital ratio will definitely fall following a rise in the profit share. This is because in the case of profit-led expansion the increase in the capital accumulation rate is sufficiently great that the effect of a lower multiplier and the possibility of a perverse effect of the implied rise in the average tax rate are wiped out. In no other case can we make a definitive statement on the effect of a rise in π on δ^* . However, if $\tau_c - \tau_w$ is negative or positive but very small and the economy is of a wage-led expansion type (which implies a_2 is small), δ^* may well increase following a rise in π .

The effect of a rise in π on growth and distribution in the long run can be seen from

$$dg^* = a_2(d\pi) + a_1\lambda(a_2 - \sigma u)(d\pi) + a_1\Lambda(d\delta^*)$$

and

$$dv^* = [v^* + q - q(i\delta/u^2)(\partial u/\partial \pi)](d\pi)/(1 - \pi) + qiu(0)(d\delta^*)/(1 - \pi)u^2.$$

In the case of profit-led expansion with $d\delta^* < 0$ the direct effect on growth is positive but the fall in δ will mitigate such a growth-stimulating effect. In the case of wage-led expansion with $d\delta^* > 0$ the possibly negative accelerator effect (the second term) is countered by the positive profitability effect (the first term) as well as the positive effect of the government debt on growth (the third term). The effect on income distribution is somewhat less ambiguous. In the case of profit-led expansion with $d\delta^* < 0$, the inequality ratio may go down despite the rise in π . But in the case of wage-led expansion with $d\delta^* > 0$ it is certain to rise.

4.4. Rise in the rate of interest

Finally, suppose that there is a rise in the rate of interest, because of a tightening of the Central Bank's monetary policy stance. This can have a variety of effects in our model. First, as is obvious from equation (1), there is a rise in consumption spending due to the higher level of interest payments by the government.¹ This has the short-run effect of increasing the rate of capacity utilisation both directly and through the accelerator-induced increase in investment demand. But this may be countered by a negative effect on consumption demand if the higher rate of interest induces capitalists to increase their savings rate by postponing current consumption, an effect ignored in our model. Second, there may be a negative effect on investment spending due to the higher cost of finance not explicitly captured in equation (2). This will tend to reduce both the rate of capacity utilisation and the rate of investment in the short run. Third, the higher rate of interest may be passed on by firms to consumers in the form of higher prices; if they do so, the mark-up rate of firms and, as a result, π will increase.² Finally, even without these effects, it will affect the rate of change in δ in the short run by increasing the expenditure side of the government budget owing to higher interest payments, as shown in equation (11).

The short- and long-run consequences of all these effects are obviously rather complicated. However, we can see that several of them have consequences very similar

¹ We assume that government bonds have a one-period life, so that a rise in the rate of interest increases interest payments for all government bonds. If we did not make this assumption, the government would have to make higher interest payments only on newly-issued bonds, and not on bonds issued earlier with fixed interest rates. Our assumption thus exaggerates the expansionary effect on consumption demand through this channel.

² See Dutt and Amadeo (1993) for an analysis of this effect in a post-Keynesian model.

to those of the other parametric shifts we have already considered: the second effect is identical to those analysed in Section 4.2, and the third effect is identical to that analysed for the rise in the profit share in Section 4.3. It thus remains to analyse the consequences of the first effect which operates *only* through the short-run equilibrium rate of capacity utilisation, and the final effect.

Consider first the consequences only of the final effect, assuming that the positive and negative effects on consumption demand offset each other exactly, causing no change in the short-run equilibrium level of u . Since we are also ignoring the investment effect, there is also no change in the short-run rate of investment. The only change in the short run is a rise in $d\delta/dt$ (as shown by equation (11)). This implies an upward shift in the curve in Fig. 1, and hence a rise in δ^* , the stable long-run equilibrium level of δ . Thus u rises in the long run, as does g . As shown in Section 3, the rise in δ in the long run increases v^* , the long-run equilibrium value of the inequality ratio; the rise in i also directly increases it, as can be seen from equation (9).

These unambiguous conclusions do not survive if we bring in the other effects. If the first effect on consumption demand reduces (increases) u in the short run, this implies that $d\delta/dt$ rises (falls) in the short run, since g also falls (rises), so that—given other things— δ^* rises (falls). Since u and g depend positively on δ , the short-run consequences can be reversed in the long run. Since the effects of a fall in investment and a rise in π have already been considered, there is no need to repeat the analysis here.

5. Government debt and income distribution

We are now in a position to answer such questions as ‘how does rising government debt affect distribution?’ or ‘what will be the distributional impact of a deficit reduction programme?’. In this section we combine the analysis of the short-run impact on Section 2, the out-of-equilibrium dynamics in Section 3 and the comparative dynamic analysis of Section 4 to answer these questions.

5.1. Expansionary fiscal policies

First, the case of a fiscal expansion through a ‘fair’ tax cut (the case of spending hike is essentially equivalent). We exclude the pervasive possibility of a fall in δ^* from our consideration and assume that δ^* rises. Suppose that there is a ‘fair’ tax cut at time t_0 from τ^*_0 to τ^*_1 (see Fig. 2). From the analysis of the short-run instantaneous impact in Section 2 we know that g jumps up and v jumps down at t_0 but that δ , being a long-run variable, cannot change instantaneously. The analysis of the out-of-equilibrium dynamics in Section 3 showed that when δ is rising, both g and v are also rising. Therefore, when the new equilibrium is reached at time t_1 , both δ^*_1 and g^*_1 must be greater than their original equilibrium values δ^*_0 and g^*_0 , but v^*_1 may be greater (Fig. 2(a)) or smaller (Fig. 2(b)) than v^*_0 as shown in our analysis in Section 4.

While it is true that a rising debt–capital ratio is associated with rising inequality (out-of-equilibrium), it is not a valid ground for opposing an expansionary fiscal policy for three reasons. As Fig. 2(b) shows, the inequality ratio may very well go down from the equilibrium level before the introduction of the tax cut owing to the initial downward jump. Moreover, it should be noted that even if the inequality ratio rises as in Fig. 2(a), it is not because of a regressive redistribution effected by financing the increased interest payments to the rich capitalists by taxes on the poor workers. In fact, the after-tax labour income is higher and grows faster in the new long-run equilibrium because of both the

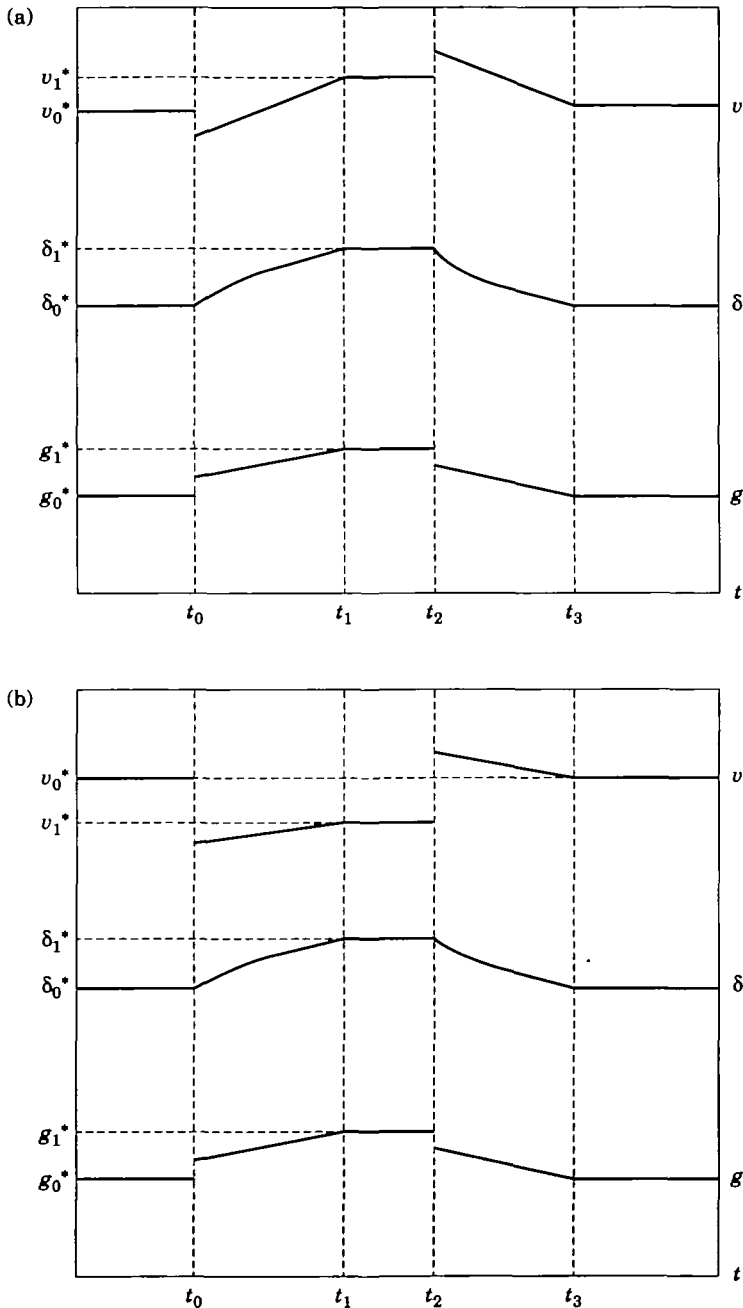


Fig. 2. Long-run effects of fiscal expansion and contraction.

lower tax rate and the rise in g . The rise in inequality is purely due to the interest income of the capitalists rising relatively faster than the produced income. Furthermore, it should also be recognised that it is quite possible that the tightening labour market conditions due to the rising growth rate may lead to a fall in the profit share of the produced income

which is certain to improve income distribution in the case of wage-led expansion and likely to do so in the case of profit-led expansion.

The argument against deficit spending on grounds of distributional equity sometimes concerns what would happen if the build-up of the debt provokes a fiscal contraction in an attempt to reduce the debt. Suppose that some political or unspecified economic constraint forces the government to restore the original debt–capital ratio δ^*_0 . So the government retracts from its initial expansionary policies and decides to raise taxes at time t_2 (Fig. 2). Clearly, the tax rates would have to be restored to the previous level τ^*_0 and the new long-run equilibrium, reached at time t_3 will have the same characteristics as the original one in terms of the debt–capital ratio, growth and distribution.¹ But immediately after the tax hike the inequality ratio will be at an all time high (even if δ^*_1 is lower than δ^*_0) as shown in Fig. 2. While this is a cause for concern and a reason to favour increasing the tax rate on the capital income rather than on the labour income, we should not lose sight of the fact that over the entire episode of expansion and contraction from time t_0 to t_3 the effect on the growth rate is favourable and the effect on the inequality ratio could also be favourable (Fig. 2(b)).

5.2. *Autonomous slowdown of growth*

When the rise in δ is caused by a fall in the autonomous investment, the stories are quite different. Figure 3 describes the case in which there is a fall in α_0 at time t_0 . The short-run impact is for g to go down and for v to go up as we saw in Section 2. From the analysis in Section 4 we know that the new equilibrium δ^*_1 reached at time t_1 must be higher than the original δ^*_0 . Therefore, during the out-of-equilibrium dynamics both g and v are also rising. Clearly, the inequality ratio is greater in the new equilibrium ($v^*_1 > v^*_0$). While it is possible that the new equilibrium growth rate g^*_1 is higher than g^*_0 , it is quite unlikely (as we discussed in Section 4) and Fig. 3 describes the case of $g^*_1 < g^*_0$.

The above analysis demonstrates that the worries about the pernicious effect of rising government debt on income distribution are much more valid when the rise in the debt is caused by an autonomous slowdown of growth rather than by expansionary fiscal policies. In this case not only is the inequality of income distribution certain to increase but, as the growth rate is almost sure to decline, it is also not misleading to claim that the rise in inequality is associated with a regressive redistribution of income from wage earners to wealthy bond holders.

As before, suppose that the government, under the pressure to reduce the debt, opts to raise taxes at time t_2 and succeeds in reducing the debt–capital ratio to δ^*_0 by the time the economy reaches the new long-run equilibrium at t_3 ($\delta^*_2 = \delta^*_0$). As shown in Fig. 3, such a debt-reduction policy will further reduce the already low growth rate ($g^*_2 < g^*_1 < g^*_0$). Income distribution could also deteriorate even more ($v^*_2 > v^*_1 > v^*_0$) as depicted in Fig. 3, although it does not necessarily do so. If this were the case, it would be a terrible irony for the left to have advocated deficit reduction out of concern for the rise in inequality from v^*_0 to v^*_1 (presumably realising and willing to accept the

¹ Baldani and Michl (1987) argue that to get back to the original deficit level after a build-up of debt due to a tax cut, the tax rate has to be higher than the level prior to the tax cut. This way they can also show that income distribution deteriorates unambiguously. But their entire analysis is based on the assumption of a fixed capital stock (while debt accumulates!), ignoring the growth effect of rising debt altogether. What is central to their claim is not the zero growth assumption itself, but its necessary corollary of the instability of the long-run equilibrium (with I fixed at the replacement of depreciating capital, $i(1 - \tau_c) > \tau\Lambda$). Once the economy moves away from an unstable equilibrium, just restoring previous τ will not bring it back to the equilibrium. One needs to take care of the accumulated debt at one shot! In this way, the leftwing argument is unwittingly connected to the rightwing unsustainability argument.

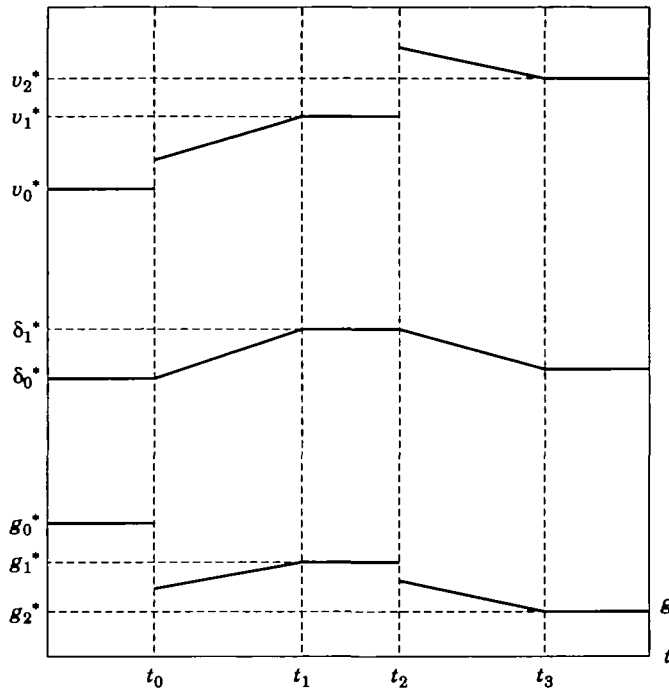


Fig. 3. Autonomous slow down of growth and debt reduction policy.

contractionary effect of deficit reduction through fiscal adjustments—traditionally an anathema to the left).

5.3. Rise in the profit share

If the debt–capital ratio rises owing to a rise in π in the case of wage-led expansion and growth, the time profile of the variables δ , g and v are exactly the same as in the case of a fall in α_0 . If the debt–capital ratio rises owing to a fall in π in the case of profit-led expansion, what will happen to g and v is ambiguous.

5.4. Rise in the rate of interest

Finally, consider the case of a rise in the rate of interest. To abstract from other issues already discussed above, consider the case in which the rise in the interest rate at t_0 which does not change the level of u in the short run, but does reduce the level of investment.

The rate of growth will fall at t_0 , while v will rise owing to the rise in i ; δ will not change in the short run. Since the long-run equilibrium value of δ is higher after the change, δ will rise over time to its long-run value δ_1^* . Meanwhile v will increase further and so will g (which may, although it is unlikely, rise above its initial long-run equilibrium level). If a debt-reduction policy is applied at t_2 , v will rise further and g will fall in the short run. Over time, δ will fall to its old value, and g and v will fall. However, since i is higher at the new long-run equilibrium value but δ is the same and fiscal parameters have changed in a contractionary direction, from (9) it follows that v must be higher than initially. Moreover, these same reasons make g lower than initially. Thus, the behaviour of the economy over time is as shown in Fig. 3.

6. Conclusion

This paper has developed a simple model of growth, distribution and government debt dynamics in a closed economy in which growth is determined along post-Keynesian lines by aggregate demand rather than by the supply of resources such as labour, and in which income is distributed between two classes—workers who earn wages and capitalists who receive profit and interest income. The main purpose of the paper was to analyse the relationship between government debt and the distribution of income.

Our analysis of this latter issue was motivated by the claims made by some economists on the left of the political spectrum that government debt worsens the distribution of income because it implies that tax revenues from workers are used to pay for the interest on the debt which is owned by the rich. Our analysis shows that this argument can be faulted on the grounds that it does not examine the possible expansionary effects of the government debt which may increase the income of workers. The precise effects on income distribution arguably depend on the reasons for a rise in the government debt, and whether these reasons create such expansionary effects. Our dynamic model was explicitly developed to analyse the dynamics of the debt–capital ratio over time and to determine what types of parametric changes could increase this ratio. We analysed the income–distributional consequences of such changes.

The main implications of our analysis may be summarised as follows.

(1) If other things are held constant, a rise in the debt–capital ratio implies an increase in income inequality. Our analysis to this extent does confirm the argument mentioned above about the growth of debt worsening the distribution of income. However, this occurs not because taxes on the poor pay for interest payments to the rich, but because interest payments on debt rise faster than after-tax income of workers which also rises owing to the expansionary effect of the rise in debt.

(2) Other things, however, are not constant. If they were, there would be no reason for the debt–capital ratio to rise if the economy were at long-run equilibrium to begin with. We examined the implications of several factors that could increase the debt–capital ratio in the long run: expansionary fiscal policies, a collapse of ‘animal spirits’, exogenous changes in the distribution of income from production (due to changes in the degree of monopoly) and a rise in the interest rate. Although in many cases we did find that the rise in the debt–capital ratio accompanied a worsening in the distribution of income, this relationship did not always hold. For instance, we found that if expansionary fiscal policies increase the debt–capital ratio it is possible for the income distribution to improve in the long run. It was also seen that income distribution necessarily improved in the short run, which means that if it took a long time to reach long-run equilibrium the distributional impact of fiscal expansion could be considered to be favourable.

(3) If the rise in the debt–capital ratio (due to any of the reasons considered above) in the long run made the government employ fiscal policies to restore the debt–capital ratio to its initial value, in addition to the unfavourable effects on growth, such policies would worsen income distribution in the short run, and quite possibly even in the long run, as our analysis of the effects of an autonomous fall in the growth rate demonstrated. Thus, efforts by the government to reduce the debt–capital ratio may well exacerbate the income distributional problem. One should thus be wary—for both growth and distributional reasons—of contractionary policies aimed at debt reduction. Moreover, we have found that one of the arguments usually made for debt reduction—the possibility of an explosively rising debt–capital ratio—appears to be invalid, since we have found that the economy will converge to a stable long-run equilibrium at some debt–capital ratio.

It should be remembered that these conclusions are provisional because they have been derived from a simple model which ignores many aspects which may be considered important for actual economies. For instance, we have ruled out the possibility of inflation, of crowding out due to increasing rates of interest, and open economy considerations. These are certainly serious omissions. But we hope that our analysis here is found useful enough to form the basis of future work which will incorporate these features into the analysis of the relationship between debt and distribution.

Bibliography

- Baldani, J. and Michl, T. R. 1987. A balanced budget multiplier for interest payments, *Journal of Post Keynesian Economics*, vol. 9, no. 3, Spring, 424–439
- Bernheim, B. D. 1989. A neoclassical perspective on budget deficits, *Journal of Economic Perspectives*, vol. 3, no. 2, Spring, 55–72
- Bhaduri, A. and Marglin, S. A. 1990. Unemployment and the real wage: the economic basis for contesting political ideologies, *Cambridge Journal of Economics*, vol. 14, no. 4, December, 375–393
- Blackburn, M. L., Bloom, D. E. and Freeman, R. B. 1990. An era of falling earnings and rising inequality? *The Brookings Review*, vol. 9, no. 1, Winter
- Blecker, R. 1989. International competition, income distribution and economic growth, *Cambridge Journal of Economics*, vol. 13, no. 3, September, 395–412
- Dornbusch, R. and Fischer, S. 1994. *Macroeconomics*, 6th edn, New York, McGraw-Hill
- Dutt, A. K. 1984. Stagnation, income distribution, and monopoly power, *Cambridge Journal of Economics*, vol. 8, no. 1, March, 25–40
- Dutt, A. K. and Amadeo, E. 1993. A post-Keynesian theory of growth, interest and money, in Baranzini, M. and Harcourt, G. C. (eds), *The Dynamics of the Wealth of Nations: Growth, Distribution and Structural Change*, New York, St. Martin's Press
- Dutt, A. K. and You, J.-I. 1993. 'The Dynamics of Government Debt and Deficits: A post-Keynesian View', unpublished manuscript, University of Notre Dame
- Economic Report of the President* 1993. Washington, DC, US Government Printing Office
- Eisner, R. 1989. Budget deficits: rhetoric and reality, *Journal of Economic Perspectives*, vol. 3, no. 2, Spring, 73–93
- Ewijk, C. van 1991. *On the Dynamics of Growth and Debt*, Oxford, Clarendon Press
- Glyn, A. 1992. The costs of stability: the advanced capitalist countries in the 1980s, *New Left Review*, September–October
- Heilbroner, R. and Bernstein, P. 1989. *The Debt and the Deficit. False Alarms/Real Possibilities*, New York, W. W. Norton
- Kalecki, M. 1971. *Selected Essays on the Dynamics of the Capitalist Economy*, Cambridge, Cambridge University Press
- Krugman, P. 1990. *The Age of Diminished Expectations*, Cambridge, MA, MIT Press
- Lavoie, M. 1992. *Foundations of Post Keynesian Economic Analysis*, Aldershot, Edward Elgar
- Michl, T. R. 1991. Debt, deficits and the distribution of income, *Journal of Post Keynesian Economics*, vol. 13, no. 3, Spring, 351–365
- National Economic Commission 1989. *Report of the National Economic Commission*, Washington, DC, GPO, March
- OECD 1993. *Employment Outlook*, Paris, OECD
- Pasinetti, L. L. 1989. Ricardian debt/taxation equivalence in the Kaldor theory of profits and income distribution, *Cambridge Journal of Economics*, vol. 13, no. 1, March, 25–36
- Phillips, K. 1990. *The Politics of the Rich and Poor*, New York, Random House
- Rock, J. M. (ed.) 1991. *Debt and the Twin Deficits Debate*, Mountain View, CA, Mayfield
- Rowthorn, R. E. 1981. Demand, real wages and economic growth, *Thames Papers in Political Economy*, Autumn, 1–39
- Steindl, J. 1952. *Maturity and Stagnation in American Capitalism*, Oxford, Blackwell
- Taylor, L. 1991. *Income Distribution, Inflation and Growth. Lectures in Structuralist Macroeconomic Theory*, Cambridge, MA, MIT Press